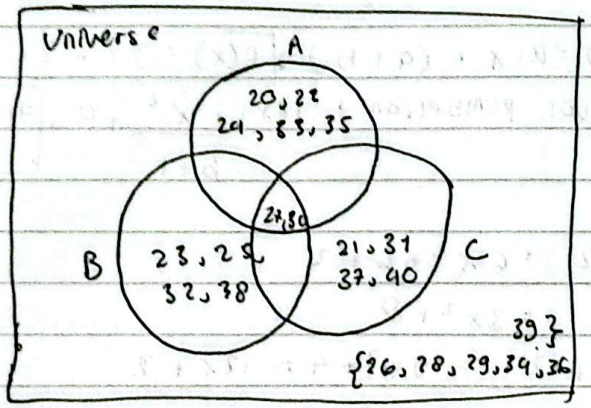
Nama: Nabilah Nur Aisyah Nugroho [SOAL NO.3] tanggal = 2026-4-21
NIM: 2510151402 6
Parameter: N = 20, a = 2, b = 0, k = 3, theta θ = 60°, alpha = 0,005

(3) dik:

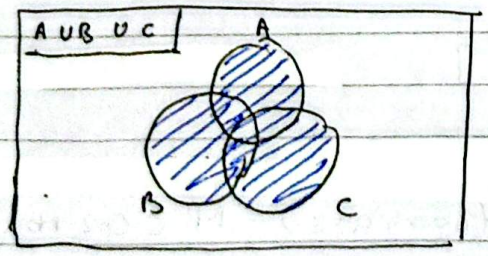
- A = {N, N+2, N+4, N+7, N+10, N+13, N+15} →
  = {20, 22, 24, 27, 30, 33, 35} (7) → media sosial
- B = {N+3, N+5, N+7, N+10, N+12, N+18}
  = {23, 25, 27, 30, 32, 38} (6) → referral
- C = {N+1, N+7, N+10, N+11, N+17, N+20}
  = {21, 27, 30, 31, 37, 40} (6) → loyalty program.
- universe = {N, N+1, ..., N+20}
  = {20, 21, ..., 40} (21)

(a) → DIAGRAM VENN

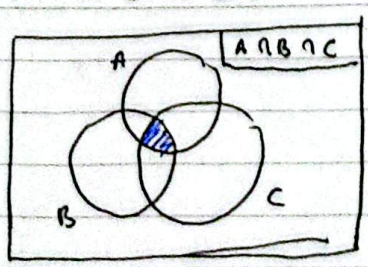


- A saja = {20, 22, 24, 33, 35} (5)
- B saja = {23, 25, 32, 38} (4)
- C saja = {21, 31, 37, 40} (4)
- A ∩ B = {27, 30} (2)
- A ∩ C = {27, 30} (2)
- B ∩ C = {27, 30} (2)
- A ∩ B ∩ C = {27, 30} (2)

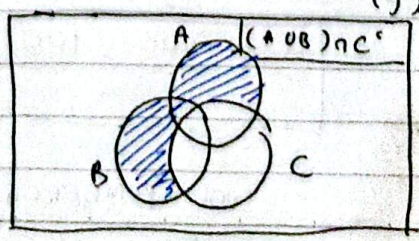
- |A ∪ B ∪ C| = |A| + |B| + |C| − |A ∩ B| − |A ∩ C| − |B ∩ C| + |A ∩ B ∩ C|
  = 7 + 6 + 6 − 2 − 2 − 2 + 2 = 15 elemen
  = {20, 21, 22, 23, 24, 25, 27, 30, 31, 32, 33, 35, 37, 38, 40}



- |A ∩ B ∩ C| = 2 = {27, 30} • |(A ∪ B) ∩ C^c| = (A ∪ B) − C



→ A ∪ B = {20, 22, 24, 27, 30, 33, 35, 23, 25, 32, 38}
→ C = {21, 27, 30, 37, 40} (buang elemen C)
→ (A ∪ B) ∩ C^c = {20, 22, 24, 33, 35, 23, 25, 32, 38}
(9) elemen



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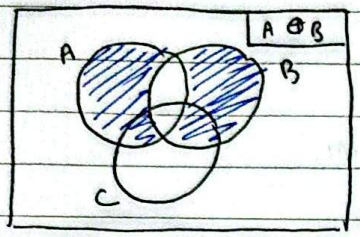
- $|A \oplus B| = (A \cup B) - (A \cap B)$
  $= \{20, 22, 24, 27, 30, 33, 35, 23, 25, 32, 38\} - \{27, 30\}$
  $= \{20, 22, 24, 33, 35, 23, 25, 32, 38\} \rightarrow 9$ elemen.



(b.) [verifikasi matlab]

(c.) dik:
fungsi harga eceran $\rightarrow f(x) = k \cdot x + (a + b) \rightarrow \boxed{f(x) = 3x + 2}$
" surcharge berdasarkan vol. pembelian $\rightarrow g(x) = \frac{x^2}{b+1} + a \rightarrow \boxed{g(x) = \frac{x^2}{1} + 2}$

Jawab:
- $(f \circ g)(x) = 3(x^2 + 2) + 2 = 3x^2 + 6 + 2$
  $= 3x^2 + 8$
- $(g \circ f)(x) = (3x + 2)^2 + 2 = 9x^2 + 4 + 12x + 2$
  $= 9x^2 + 12x + 6$
- apakah komutatif ? buktikan pakai $x = b + 2 \rightarrow x = 0 + 2 \rightarrow \boxed{x = 2}$

$\rightarrow (f \circ g)(x) = f(g(2)) = 3(2^2) + 8 = 3(4) + 8 = 20$
$(g \circ f)(x) = g(f(2)) = 9(2^2) + 12(2) + 6 = 36 + 24 + 6 = 66$
$f(g(2)) = 20 \neq 66 = g(f(2))$
[TERBUKTI, TIDAK KOMUTATIF.]

(d.) dik:
, pendapatan 1 pelanggan = Rp $(k \cdot 100 + a + b)$ = Rp $(300 + 0 + 2)$ = Rp 302 ,-
$\rightarrow f(x) = 3x + 2$
$\rightarrow f^{-1}(x) = ?$ (vol. pembelian yg menghasilkan)

Jawab:
$f^{-1}(x) =$
$y = 3x + 2$
$y - 2 = 3x$
$\frac{y-2}{3} = x$

$\boxed{f^{-1}(x) = \frac{x-2}{3}}$

$\rightarrow x = f^{-1}(302)$
$= \frac{302 - 2}{3}$
$\boxed{x = \frac{300}{3} = 100}$ //

vol. pembelian yg menghasilkannya C adalah 10 C.

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(e.) diu :

$f(x) = 3x + 2$

$g(x) = x^2 + 2$

domain = R, kodomain = R

dit :

klasifikasikan fungsi f & g : injektif ? surjektif ? bijektif ?

$f(x) = 3x + 2$

→ Injektif ?

definisi : $f(x_1) = f(x_2) \rightarrow x_1 = x_2$

Bukti :

$$3x_1 + 2 = 3x_2 + 2$$
$$3x_1 = 3x_2$$
$$x_1 = x_2$$

✓ injektif

→ Surjektif ?

tiap $y \in R$, ada x sehingga $f(x) = y$

Bukti :

$$y = 3x + 2$$
$$x = \frac{y-2}{3}$$

karena $x \in R$, selalu ada solusi

✓ surjektif

KESIMPULAN $f(x) = 3x + 2$
✓ injektif
✓ surjektif
✓ bijektif.

$g(x) = x^2 + 2$

→ Injektif ?

bukti :

$g(1) = 3$, $g(-1) = 3$

$g(1) \neq g(-1)$

$1 \neq -1$

✗ injektif

→ Surjektif ?

range → $g(x) = x^2 + 2 \geq 2$

→ range = $[2, \infty)$

tapi kodomain = R

$y = 0$ → tidak ada

✗ surjektif

KESIMPULAN $g(x) = x^2 + 2$
✗ injektif
✗ surjektif
✗ bijektif

PAPERLINE